Fourteen initial Θ-data on one curve over $\mathbb{Q}(i)$

Toshihisa Urahigashi
ORCID 0009-0004-0460-6242
123026.urahigashi.toshihisa@gmail.com

Version 1.0.0, 6 November 2026
doi:10.5281/zenodo.23990717

Abstract

Let $E\colon y^2 = x(x-1)(x-a)$ with $a = (41+26i)^{472}$, a curve over $\mathbb{Q}(i)$. We prove that for each prime ℓ in

$$\Lambda = \{89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157\}$$

the curve extends to an initial Θ-datum in the sense of [IUT1, Def. 3.1], and that this datum satisfies the hypotheses of [IUT4, Thm. 1.10]. The field of moduli is $\mathbb{Q}(i)$. Above the split prime 2357 there is one place of multiplicative and one of good reduction, with ramification indices 15ℓ and 1. The curve and the fourteen levels come from [Second, §13]. That report showed that the levels pass five numerical conditions, and said explicitly that they had not been verified to be initial Θ-data; the present report gives the verification. The proof uses the individual ℓ only through five integer facts and one Frobenius witness per level. The norm of $1-a$ has 1592 digits, and its factorization is not used. Instead, the valuation of $1-a$ at each split place is bounded through the cyclotomic factor $\Phi_d(\pi)$, where d is the order of π modulo the place, and the 138,731 primes that this bound does not exclude are tested directly; the inert places and the places above 2357 are treated directly. The fourteen levels are the primes in $[\sqrt{\log \mathfrak{q}^{\forall}}, 472/3)$, and they satisfy condition (C1) of [IUT4, Cor. 2.2]. Nothing is claimed about [IUT3, Cor. 3.12] or about *abc*.

1 Introduction

[S] An initial Θ-datum [IUT1, Def. 3.1] is a list of objects — a number field F, a once-punctured elliptic curve X_F over F, a prime ℓ, an orbicurve $\underline{C}_K$, a set $\underline{\mathbb{V}}$ of places of a field K, a set $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ of places of the field of moduli, and a cusp $\underline{\epsilon}$ — subject to conditions (a)–(f). Dupuy and Hilado work out the curve 11a1 over $\mathbb{Q}$ [DH1, §2.10], and Zhou describes constructions of μ_6-initial Θ-data over $\mathbb{Q}$ [Zhou1, Prop. 2.5, Lemma 2.6]. Over quadratic fields, [First] gives a datum over $\mathbb{Q}(\sqrt{5})$ and [Second] one over $\mathbb{Q}(i)$, both at $\ell = 7$, and the catalogue [Cat] lists these and several others.

[S] In [Second, §13] the curve of this report appears with the fourteen levels of Λ, and two parts of condition (c) are settled there: $\ell \nmid \mathrm{N}(1-a)$, and the sufficient condition for the order condition that $\mathrm{N}(1-a)$ is free of 89th powers. The same section says that the fourteen levels are *not* verified initial Θ-data: the argument for the Galois image had been carried out only for the datum at $\ell = 7$, and conditions (a)–(f) had not been checked at these levels. In [Cat] the curve is listed as $D^{(\ell)}_{2357}$, with its values marked as conditional on such a verification. [P] This report supplies it:

- the image condition at each level, from a transvection at the bad place above 2357 and one Frobenius witness per level (§4.3);

- the order condition at every selected place, by an argument that does not use the factorization of $1-a$ and that is independent of the computation of [Second, §13] (Proposition 3.2(d));
- the remaining conditions of [IUT1, Def. 3.1] and the hypotheses of [IUT4, Thm. 1.10], by the arguments of [Second, §§3–4] with ℓ in place of 7 (§4).

[P] *Scope.* This report constructs data and verifies their defining conditions. Beyond the local invariants of Theorem 2.1(iii) and the margin quoted in Remark 5.1, it does not evaluate quantities of [IUT3, Cor. 3.12] or [IUT4, Thm. 1.10] at these data, and it makes no claim about either statement or about *abc*. Each statement is marked [S] if it is a statement of a cited source, paraphrased with its locator, and [P] if it is proved or computed here.

2 The datum and the theorem

$$\begin{aligned} &L=\mathbb{Q}(i), && \pi=41+26i, \quad \mathrm{N}(\pi)=2357, \quad a=\pi^{472},\\ &E\colon y^2=x(x-1)(x-a), && \ell\in\Lambda, \\ &F=L(E[15]), && K=F(E[\ell])=L(E[15\ell]). \end{aligned} \tag{1}$$

Let $X_F=E_F\setminus\{0\}$ and $C_F=[X_F/\{\pm1\}]$. Take as $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ the set of places of L of odd residue characteristic at which E has multiplicative reduction. By §4.1 below, these are the place v_π and the places of odd residue characteristic dividing $1-a$; by Proposition 3.2(e) none of them lies above ℓ, so the same set is obtained by taking the places of multiplicative reduction outside 2ℓ, as in [Note, §2] and [Cat, §3].

Theorem 2.1. [P] *Let $\ell\in\Lambda$.*

(i) The data (1) *extend to an initial Θ-datum $(\overline{F}/F, X_F, \ell, \underline{C}_K, \underline{\mathbb{V}}, \mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}, \underline{\epsilon})$ in the sense of [IUT1, Def. 3.1], with $F_{\mathrm{mod}}=L$.*

(ii) This datum satisfies the hypotheses of [IUT4, Thm. 1.10]: E_F has good reduction at every nonarchimedean place of F that lies over $\mathbb{V}_{\mathrm{mod}}\setminus\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ and does not divide 2ℓ, and $F=F_{\mathrm{mod}}(\sqrt{-1}, E_{F_{\mathrm{mod}}}[2\cdot3\cdot5])$ in the sense stated there.

(iii) Above 2357 there are exactly two places of L, the place $v_\pi\in\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ and the place $v_{\bar\pi}$ of good reduction. At the places of $\underline{\mathbb{V}}$ above them, the ramification indices of K over $\mathbb{Q}_{2357}$ are

$$e_{\mathrm{b}}=15\ell, \qquad e_{\mathrm{g}}=1,$$

and $\mathrm{ord}_{2357}(\underline{q}_{\mathrm{b}})=472/\ell$, where $\underline{q}_{\mathrm{b}}$ is the 2ℓ-th root of the q-parameter of [IUT1, Example 3.2(iv)]. In particular $e\le 2357-2=2355$ at both places, and $e_{\mathrm{b}}=2355$ exactly when $\ell=157$.

Remark 2.2 (one curve, fourteen levels)**.** [P] The field of moduli, the curve, the prime 2357 and the set $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ do not depend on ℓ; only ℓ and, with it, K, the covering, $\underline{\mathbb{V}}$ and the cusp change. The proof of (i) and (ii) uses the individual level only through the five integer facts

$$\ell\neq 2357, \qquad \ell\nmid 944, \qquad \ell\nmid \mathrm{N}(1-a), \qquad \ell\nmid 23040, \qquad \mathrm{ord}_{\mathfrak{p}}(1-a)\le 88<\ell,$$

and through one Frobenius witness from Table 2; part (iii) also uses $\ell\le 157$. The set Λ is taken from [Second, §13], where it is the set of primes in a window; the window plays no role in the proof (see §5).

3 The arithmetic of a and $1-a$

[P] The order condition at the bad places dividing $1-a$ rests on the following elementary bound.

Lemma 3.1. [P] *Let $\mathfrak{q}$ be a prime of $\mathbb{Z}[i]$ with $\mathfrak{q} \neq (\pi)$, whose residue characteristic p is neither 2 nor 59, and suppose that $\mathfrak{q} \mid \pi^{472}-1$. Let d be the order of π in $(\mathbb{Z}[i]/\mathfrak{q})^\times$. Then $d \mid 472$, $d \mid \mathrm{N}\,\mathfrak{q}-1$, and*

$$\mathrm{ord}_{\mathfrak{q}}(\pi^{472}-1) = \mathrm{ord}_{\mathfrak{q}}(\Phi_d(\pi)).$$

In particular $(\mathrm{N}\,\mathfrak{q})^k \leq \mathrm{N}(\Phi_d(\pi))$ for $k = \mathrm{ord}_{\mathfrak{q}}(\pi^{472}-1)$.

Proof. The group $(\mathbb{Z}[i]/\mathfrak{q})^\times$ has order $\mathrm{N}\,\mathfrak{q}-1$, and $\pi^{472} \equiv 1$, so d divides both. Put $x = \pi^d$ and $m = 472/d$. Then $\pi^{472}-1 = (x-1)(1+x+\cdots+x^{m-1})$, and the second factor is congruent to m modulo $\mathfrak{q}$, a unit because m divides $2^3 \cdot 59$ and $p \nmid 2 \cdot 59$. So $\mathrm{ord}_{\mathfrak{q}}(\pi^{472}-1) = \mathrm{ord}_{\mathfrak{q}}(\pi^d-1)$. Next, $\pi^d - 1 = \prod_{e \mid d} \Phi_e(\pi)$, and for $e \mid d$, $e < d$, the factor $\Phi_e(\pi)$ divides $\pi^e - 1$, which is prime to $\mathfrak{q}$ by the minimality of d. So $\mathrm{ord}_{\mathfrak{q}}(\pi^d-1) = \mathrm{ord}_{\mathfrak{q}}(\Phi_d(\pi))$. Finally, if $\mathfrak{q}^k$ divides the nonzero element $\Phi_d(\pi)$, then $(\mathrm{N}\,\mathfrak{q})^k$ divides $\mathrm{N}(\Phi_d(\pi))$. □

d	$\deg \Phi_d$	digits of $\mathrm{N}(\Phi_d(\pi))$	$R_d = \lfloor \mathrm{N}(\Phi_d(\pi))^{1/89} \rfloor$	$\mathrm{lcm}(4,d)$
1, 2, 4, 8	1, 1, 2, 4	4, 4, 7, 14	1	—
59, 118	58	196	157	236
236	116	392	24856	236
472	232	783	617831325	472

Table 1: The cyclotomic factors $\Phi_d(\pi)$, $d \mid 472$. Each is computed from π by exact division, $\Phi_d(\pi) = \prod_{e \mid d} (\pi^e - 1)^{\mu(d/e)}$, and their product is checked to equal $\pi^{472}-1$.

Proposition 3.2. [P]

(a) $j(E) = 256(a^2-a+1)^3/(a^2(a-1)^2)$ is not real. Hence $j(E) \notin \mathbb{Q}$ and $F_{\mathrm{mod}} = \mathbb{Q}(j(E)) = L$.

(b) 2357 is prime, $2357 = \pi\bar{\pi}$ splits in L, and

$$\mathrm{ord}_{\pi}(a) = 472, \qquad \mathrm{ord}_{\bar{\pi}}(a) = 0, \qquad \mathrm{ord}_{\pi}(1-a) = \mathrm{ord}_{\bar{\pi}}(1-a) = 0.$$

(c) $\mathrm{N}(1-a)$ is an integer of 1592 digits with $v_2 = 9$, $v_3 = 2$, $v_5 = 2$, and $\gcd(\mathrm{Re}(1-a), \mathrm{Im}(1-a)) = 240$.

(d) At every place $\mathfrak{p}$ of L of odd residue characteristic, $\mathrm{ord}_{\mathfrak{p}}(1-a) \leq 88$.

(e) For every $\ell \in \Lambda$: $\ell \neq 2357$, and ℓ divides none of $944 = 2^4 \cdot 59$, $23040 = 2^9 \cdot 3^2 \cdot 5$ and $\mathrm{N}(1-a)$.

Proof. (a) Write $j = A/B$ with $A = 256(a^2-a+1)^3$ and $B = a^2(a-1)^2$ in $\mathbb{Z}[i]$. The imaginary part of $A\bar{B}$ is a nonzero integer (of 7953 digits after removing the common factor with $\mathrm{N}(B)$), so $j \neq \bar{j}$. As a second check, at the two places of L above 29 ($i \equiv 12$ and $i \equiv 17$) the residue of a is 20 and 23, and j reduces to 16 and 18 in $\mathbb{F}_{29}$; a rational j would reduce to the same element at both. The field of moduli of an elliptic curve is generated by its j-invariant [IUT4, Prop. 1.8(ii)], and $j \in L \setminus \mathbb{Q}$.

(b) $\bar{\pi}$ is not an associate of π. Modulo π one has $a \equiv 0$, so $1-a \equiv 1$. Modulo $\bar{\pi}$ one has $i \equiv 41 \cdot 26^{-1}$ and $\pi \equiv 82$; since $82^4 \equiv 202 \not\equiv 1 \pmod{2357}$ and $\gcd(472, 2356) = 4$, $82^{472} \not\equiv 1$, so $\bar{\pi} \nmid 1-a$.

(c), (e) Exact computation (§6); $\mathrm{N}(1-a) \bmod \ell$ was also recomputed in $\mathbb{F}_\ell[i]$ from π.

(d) If $\mathfrak{p} = (p)$ is inert, then $\mathrm{ord}_{\mathfrak{p}}(1-a)$ is the smaller of $v_p(\mathrm{Re}(1-a))$ and $v_p(\mathrm{Im}(1-a))$, so $\mathrm{ord}_{\mathfrak{p}}(1-a) \le v_p(240) \le 1$; this covers $p = 3$ and $p = 59$. The places above 2357 are covered by (b). The remaining places are the places $\mathfrak{q}$ of degree one above the primes $p \equiv 1 \pmod 4$, $p \neq 2357$, and for them we use Lemma 3.1 and Table 1. Suppose $\mathrm{ord}_{\mathfrak{q}}(1-a) \ge 89$, and let d be the order of π modulo $\mathfrak{q}$. By the lemma, $p = \mathrm{N}\,\mathfrak{q} \le R_d = \lfloor \mathrm{N}(\Phi_d(\pi))^{1/89} \rfloor$, and $p \equiv 1 \pmod{\mathrm{lcm}(4,d)}$. For $d \mid 8$ this is impossible, since $R_d = 1$. For $d \in \{59, 118\}$ it is impossible as well, since $R_d = 157$ and no $p \equiv 1 \pmod{236}$ is at most 157. So p is a prime $p \equiv 1 \pmod{236}$ with $p \le 24856$ or a prime $p \equiv 1 \pmod{472}$ with $p \le 617{,}831{,}325$. There are 138,731 such primes, and at each of their 277,462 places the valuation of $1-a$ was computed. Exactly four of these places divide $1-a$, one above each of $p = 1181$, 21713, 2876369 and 6188393, and each divides it exactly once. Hence no place has $\mathrm{ord}_{\mathfrak{q}}(1-a) \ge 89$. □

Remark 3.3 (a second computation)**.** [S] [Second, §13] proves the stronger statement that $\mathrm{N}(1-a)$ is free of 89th powers. It splits Φ_{472} over $\mathbb{Z}[i]$ into two factors of degree 116, sieves each of the nine resulting block norms up to its own 89th root (at most 24856; 8318 primes in all), and checks with the pairwise greatest common divisors that only 2 and 5 divide two blocks. Its program is in the deposit of [Second]. [P] The computation above does not split Φ_{472} and shares no code with it; both run in seconds and agree. A direct search for 89th powers in $\mathrm{N}(1-a)$ would need the primes up to $\mathrm{N}(1-a)^{1/89} \approx 10^{17.9}$.

4 Proof of Theorem 2.1

4.1 Conditions (a) and (b)

[S] Condition (a) of [IUT1, Def. 3.1] asks that F be a number field with $\sqrt{-1} \in F$. Condition (b) asks that X_F have stable reduction at every nonarchimedean place of F; that $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ be a nonempty set of nonarchimedean places of F_{mod} of odd residue characteristic such that X_F has bad multiplicative reduction at every place of F above $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$; that F/F_{mod} be Galois of degree prime to ℓ; and that the $2 \cdot 3$-torsion points of E_F be rational over F.

[P] $i \in L \subseteq F$. The field of moduli of X_F is that of E, which is L by Proposition 3.2(a). The extension $F = L(E[15])$ is Galois over L, and its group embeds in $\mathrm{GL}_2(\mathbb{Z}/15\mathbb{Z})$, of order $48 \cdot 480 = 23040 = 2^9 \cdot 3^2 \cdot 5$, so $[F : L]$ is prime to ℓ (Proposition 3.2(e)). The 2-torsion of E is rational over L (Legendre form), and the 3-torsion is rational over F since $E[3] \subseteq E[15]$.

[S] By [IUT4, Prop. 1.8(v)], if k is a perfect complete discretely valued field, $l \ge 3$ is a prime invertible in its ring of integers and all l-torsion points of an elliptic curve E_k are k-rational, then E_k has semistable reduction. [P] At a completion F_w apply this with $l = 3$ if the residue characteristic is not 3, and with $l = 5$ otherwise. So E_F has semistable reduction at every nonarchimedean place: good where $\mathrm{ord}(j) \ge 0$ and multiplicative where $\mathrm{ord}(j) < 0$ [Sil, Props. VII.5.1, VII.5.5]. At a place of good reduction the minimal model is smooth, and at a place of multiplicative reduction its special fibre is a nodal cubic on which the origin is a smooth point; in both cases X_F has stable reduction.

[P] The discriminant of the model (1) is $16a^2(a-1)^2$, and $c_4 = 16(a^2 - a + 1)$. At an odd place where a and $1-a$ are units, the discriminant is a unit and E has good reduction. At v_π, and at an odd place $\mathfrak{p}$ dividing $1-a$, one has $a^2 - a + 1 \equiv 1$, so c_4 is a unit, the model is minimal [Sil, §VII.1], and the reduction is multiplicative, with

$$\mathrm{ord}_\pi(j) = -944, \qquad \mathrm{ord}_{\mathfrak{p}}(j) = -2\,\mathrm{ord}_{\mathfrak{p}}(1-a).$$

So $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ consists of v_π and the places of odd residue characteristic dividing $1-a$; it is nonempty, and its elements have odd residue characteristic. Since $\mathrm{ord}(j) < 0$ and the reduction over F is semistable, the reduction is multiplicative at every place of F above $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$.

4.2 Condition (c): residue characteristics and orders

[S] Condition (c) asks that $\ell \geq 5$ be a prime; that the image of the outer homomorphism $G_F \to \mathrm{GL}_2(\mathbb{F}_\ell)$ determined by the ℓ-torsion points of E_F contain $\mathrm{SL}_2(\mathbb{F}_\ell)$, K being the field cut out by its kernel; and that ℓ be prime to the residue characteristics of the places in $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ and to the orders of the q-parameters of E_F at the places of F above $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ (the "local heights" of [GenEll, Def. 3.3]).

[P] The residue characteristics of $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ are 2357 and the odd primes dividing $\mathrm{N}(1-a)$, and ℓ is none of them (Proposition 3.2(e)). The orders of the q-parameter are $\mathrm{ord}_\pi(q) = -\mathrm{ord}_\pi(j) = 944$ at v_π and $\mathrm{ord}_\mathfrak{p}(q) = 2\,\mathrm{ord}_\mathfrak{p}(1-a)$ at an odd place $\mathfrak{p} \mid 1-a$, where $1 \leq \mathrm{ord}_\mathfrak{p}(1-a) \leq 88$ by Proposition 3.2(d). At a place w of F above v the order is multiplied by $e(w/v)$, which divides $[F : L] \mid 23040$. As ℓ is a prime with $\ell \nmid 944$, $\ell \nmid 23040$ and $\ell \geq 89$, it divides none of these orders.

4.3 Condition (c): the image

Lemma 4.1. [P] *Let ℓ be a prime and let $H \subseteq G \subseteq \mathrm{GL}_2(\mathbb{F}_\ell)$ be subgroups with H normal in G. Suppose that H contains a transvection $T \neq 1$ and that G contains an element g whose characteristic polynomial is irreducible over $\mathbb{F}_\ell$. Then $H \supseteq \mathrm{SL}_2(\mathbb{F}_\ell)$.*

Proof. Let $D = \ker(T-1)$, a line. Since the characteristic polynomial of g has no root in $\mathbb{F}_\ell$, g has no eigenvector, so $gD \neq D$. Then $T' = gTg^{-1}$, which lies in H, is a transvection $\neq 1$ with fixed line gD. In a basis $e_1 \in D$, $e_2 \in gD$ the two read $\left(\begin{smallmatrix}1 & x\\ 0 & 1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1 & 0\\ y & 1\end{smallmatrix}\right)$ with $x, y \neq 0$. Since ℓ is prime, their powers are all the matrices $\left(\begin{smallmatrix}1 & t\\ 0 & 1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1 & 0\\ t & 1\end{smallmatrix}\right)$, $t \in \mathbb{F}_\ell$, and these generate $\mathrm{SL}_2(\mathbb{F}_\ell)$. □

[P] We apply the lemma with H and G the images of $\mathrm{Gal}(\overline{\mathbb{Q}}/F)$ and $\mathrm{Gal}(\overline{\mathbb{Q}}/L)$ in $\mathrm{GL}(E[\ell]) \simeq \mathrm{GL}_2(\mathbb{F}_\ell)$; H is normal in G because F/L is Galois.

[P] *The transvection.* Let w be a place of F above v_π. Over F_w the curve is a split Tate curve (§4.4; [Sil, App. C, Thm. 14.1]), with $\mathrm{ord}_w(q) = e(w/v_\pi) \cdot 944$, and this is prime to ℓ by §4.2. As $\ell \neq 2357$, the maximal unramified extension of F_w contains μ_ℓ, and $q^{1/\ell}$ generates a ramified extension of it of degree ℓ. So an element of the inertia group at w acts on $E[\ell] = \langle \zeta_\ell, q^{1/\ell} \rangle$ by a transvection $T \neq 1$, and $T \in H$.

[P] *The Frobenius witnesses.* For a prime $q \equiv 1 \pmod 4$ and a root r of $r^2 \equiv -1 \pmod q$, the place $\mathfrak{q} = (q, i - r)$ of L sends a to $\mathrm{Re}(a) + r\,\mathrm{Im}(a) \bmod q$. At the four places of Table 2 this residue is neither 0 nor 1, so E has good reduction there. The point count gives $a_\mathfrak{q} = q+1-\#E(\mathbb{F}_q)$. For $\ell \neq q$, a Frobenius element $g \in G$ at $\mathfrak{q}$ acts on $E[\ell] \simeq \tilde{E}[\ell]$ [Sil, Prop. VII.3.1] with characteristic polynomial $X^2 - a_\mathfrak{q} X + q$ [Sil, Thm. V.2.3.1], which is irreducible over $\mathbb{F}_\ell$ exactly when $a_\mathfrak{q}^2 - 4q$ is a nonsquare modulo ℓ.

q	$i \equiv$	$a \equiv$	$\#E(\mathbb{F}_q)$	$a_\mathfrak{q}$	$a_\mathfrak{q}^2 - 4q$	levels covered
13	5	3	16	-2	$-48 = -3 \cdot 4^2$	89, 101, 107, 113, 131, 137, 149
17	13	16	16	2	$-64 = -8^2$	103, 107, 127, 131, 139, 151
29	12	20	32	-2	$-112 = -7 \cdot 4^2$	89, 97, 101, 103, 131, 139, 157
97	75	75	96	2	$-384 = -6 \cdot 8^2$	89, 109, 113, 137, 139, 157

Table 2: Frobenius witnesses. The last column lists the $\ell \in \Lambda$, $\ell \neq q$, at which $a_\mathfrak{q}^2 - 4q$ is a nonsquare modulo ℓ. Every $\ell \in \Lambda$ occurs in at least one row.

[P] By quadratic reciprocity, the first three rows are the conditions $\ell \equiv 2 \pmod 3$, $\ell \equiv 3 \pmod 4$ and $(\ell/7) = -1$. The fourth, $(-6/\ell) = -1$, is needed only at $\ell = 109$, which no other row covers. The choice of place matters: above 17 the other place ($i \equiv 4$) sends a to 1 and

belongs to $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$, and above 97 the other place ($i \equiv 22$) has $\#E(\mathbb{F}_{97}) = 112$, $a_{\mathfrak{q}} = -14$ and discriminant -192.

[P] So, for every $\ell \in \Lambda$, Lemma 4.1 applies with the transvection at w and a witness from Table 2, and the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/F)$ contains $\mathrm{SL}_2(\mathbb{F}_\ell)$.

4.4 Split reduction, and the local invariants above 2357

[P] At an odd place where $a \equiv 0$, the special fibre of (1) is $y^2 = x^2(x-1)$, with tangent cone $y^2 = -x^2$ at the node; it splits because $i \in L$. At an odd place where $a \equiv 1$, putting $t = x - 1$ gives $y^2 = (1+t)t^2$, with tangent cone $y^2 = t^2$, already split. So the reduction is split multiplicative over L at every place of $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$, and it stays split over F and K.

[P] At $v_{\bar\pi}$ the reduction is good and 15ℓ is prime to 2357, so K is unramified there and $e_{\mathrm{g}} = 1$. At v_π the curve is a split Tate curve over $\mathbb{Q}_{2357}$ with $\mathrm{ord}_\pi(q) = 944$. Adjoining $E[15\ell]$ adjoins $\mu_{15\ell}$, which is unramified, and a 15ℓ-th root of q; since $\gcd(15\ell, 944) = 1$, the ramification index is exactly 15ℓ. So $e_{\mathrm{b}} = 15\ell$ and $\mathrm{ord}_{2357}(q_{\underline{\underline{\mathrm{b}}}}) = 944/(2\ell) = 472/\ell$. Finally $15\ell \le 15 \cdot 157 = 2355 = 2357 - 2$, with equality only at $\ell = 157$.

4.5 Condition (d): the quotient and the core

[P] Write $V = E[\ell](K)$, fix a line $W \subset V$ and an element $0 \neq \xi \in Q = V/W$, and let $\pi_W \colon E_K \to E' = E_K/W$, with $\psi \colon E' \to E_K$ such that $\psi\pi_W = [\ell]$. Restriction gives an isomorphism $\bar\pi_W \colon Q \simeq \ker\psi$, so this kernel is a constant group over K. Put $\underline{X}_K = E' \setminus \ker\psi$, $\underline{C}_K = [\underline{X}_K/\{\pm1\}]$ and $C_K = [X_K/\{\pm1\}]$. The covering $\underline{X}_K \to X_K$ is Galois with group $\ker\psi \simeq Q$. It extends to the isogeny ψ of the compactifications, so it is unramified above the cusp, and its fibre there, $\ker\psi$, consists of K-rational points; so the decomposition group of the cusp maps trivially to Q. Passing to the quotients by ±1, with the cusp 0 of $\underline{X}_K$ as origin, gives an orbicurve of type $(1, \ell\text{-tors})_\pm$ in the sense of [EtTh, Def. 2.1]. Let $\underline{\epsilon}$ be the cusp corresponding to $\pm\bar\pi_W(\xi)$.

[S] By [MFHMP, Prop. 2.1], if the once-punctured curve of an elliptic curve over a field of characteristic zero is arithmetic, then its j-invariant is one of $2^{14} \cdot 31^3 \cdot 5^{-3}$, $2^2 \cdot 73^3 \cdot 3^{-4}$, 1728, 0. [P] Here $j(E)$ is not real (Proposition 3.2(a)), so $X_{\overline{K}}$ is not arithmetic. The finite étale map $X_{\overline{K}} \to C_{\overline{K}}$ puts the two orbicurves in the same commensurability class, so $C_{\overline{K}}$ is not arithmetic either. [S] A punctured hemi-elliptic orbicurve over an algebraically closed field of characteristic zero that is not arithmetic is a core [CanLift, Prop. 2.7]; an orbicurve over k is a k-core if and only if its base change to a separable closure is a core there [CanLift, Prop. 2.3(i)]; and the property is unchanged under extension of algebraically closed fields of characteristic zero [CanLift, Prop. 2.3(ii)]. [P] So C_K is a K-core, and, since $\underline{C}_K \to C_K$ is finite étale, it is the K-core of $\underline{C}_K$ required by [IUT1, Def. 3.1(d)]. For a completion K_v, extend $K \hookrightarrow \overline{K}$ to an embedding of $\overline{K}$ into an algebraic closure of K_v and apply [CanLift, Prop. 2.3(ii)] and then [CanLift, Prop. 2.3(i)]; no real completion occurs, since $i \in K$. The open subgroups $\Delta_{\underline{\underline{X}}} \subseteq \Delta_{\underline{\underline{C}}} \subseteq \Delta_{\underline{C}}$ of [IUT1, Def. 3.1(d)] are supplied by [EtTh, Prop. 2.2, Def. 2.3]. Since $\mathrm{ord}_\pi(j) = -944 < 0$, j is not an algebraic integer and E has no complex multiplication; this is not used above.

4.6 Conditions (e) and (f): the places and the cusp

[S] Condition (e) asks for a subset $\underline{\mathbb{V}} \subseteq \mathbb{V}(K)$ mapping bijectively to $\mathbb{V}(F_{\mathrm{mod}})$, such that at every $\underline{v} \in \underline{\mathbb{V}}$ above $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ the orbicurve $\underline{C}_{\underline{v}}$ is of type $(1, \mathbb{Z}/\ell\mathbb{Z})_\pm$ [EtTh, Def. 2.5(i)]. That definition makes two assumptions: that the quotient $\overline{\Pi}^{\mathrm{ell}}_X \twoheadrightarrow Q$ factor through the natural quotient $\Pi_X \twoheadrightarrow \widehat{\mathbb{Z}}$ determined by the quotient $\Pi^{\mathrm{tp}}_X \twoheadrightarrow \mathbb{Z}$ of [EtTh, §1], and that the splitting of $\overline{D}_x \twoheadrightarrow G_K$ which determines the covering $\underline{\underline{X}} \to \underline{X}$ be compatible with the $\{\pm1\}$-structure of [EtTh, Thm. 1.10(iii)], which is defined when the residue characteristic is odd. Condition (e)

records that the 2-torsion part of (b) gives the condition "$K = \ddot{K}$" of [EtTh, Def. 2.5(i)], and that the models of type $(1, (\mathbb{Z}/\ell\mathbb{Z})^{\Theta})$ and $(1, (\mathbb{Z}/\ell\mathbb{Z})^{\Theta})_{\pm}$ at such places are supplied by the theory of [EtTh, §2]. Condition (f) asks that $\underline{\epsilon}$ arise from a nonzero element of Q, and that at each $\underline{v} \in \underline{\mathbb{V}}$ above $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ its localization arise, up to sign, from the canonical generator of the quotient $\widehat{\mathbb{Z}}$ of [EtTh, Def. 2.5(i)].

[P] At each $u \in \mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ choose a place of F above u and a place w of K above it. The split Tate sequence $0 \to \mu_\ell \to E(K_w)[\ell] \to \mathbb{Z}/\ell \to 0$, in which the class of $q^{1/\ell}$ maps to 1, determines a line $W_w = \mu_\ell \subset V$ and an element $\pm\xi_w$ of V/W_w, up to sign: changing the root $q^{1/\ell}$ changes it by an element of $\mu_\ell = W_w$, which does not change its class, and reversing the orientation of the dual graph changes the sign. The group $\mathrm{SL}(V)$ acts transitively on such pairs: for a pair $(W', \pm\xi')$ choose $s \in W'$ and a lift t of ξ' with $\det(s,t) = 1$, and note that $\mathrm{SL}(V)$ acts transitively on the bases so obtained. So, by §4.3, there is $\sigma_u \in \mathrm{Gal}(K/F)$ with $\sigma_u(W_w, \pm\xi_w) = (W, \pm\xi)$. Put $v(u) = \sigma_u w$, the place with $|b|_{\sigma_u w} = |\sigma_u^{-1} b|_w$; the isomorphism $\sigma_u^{-1}\colon K_{v(u)} \simeq K_w$ commutes with localization, so the fixed global pair $(W, \pm\xi)$ acquires at $v(u)$ the local marking. The global W and $\underline{\epsilon}$ are not changed from place to place. Since σ_u fixes F, the place $v(u)$ lies above the same u, and the fibres above different u are disjoint, so these choices can be made simultaneously. Choosing an arbitrary place of K above each remaining place of F_{mod} gives $\underline{\mathbb{V}} \xrightarrow{\sim} \mathbb{V}(F_{\mathrm{mod}})$.

[P] Locally at $v(u)$ the quotient is $E(q) \to E(q^\ell)$, $z \mapsto z^\ell$, and its dual is $E(q^\ell) \to E(q)$, $z \mapsto z$, which multiplies the period of the dual graph by ℓ. So the quotient $\Pi_X \to Q$ factors through $\widehat{\mathbb{Z}} \to \mathbb{Z}/\ell$, which is the first assumption of [EtTh, Def. 2.5(i)]. The second concerns only the splitting that determines $\underline{\underline{X}}_{v(u)}$; we take it to be the one of the models supplied by [EtTh, §2] and recorded in condition (e). So $\underline{C}_{v(u)}$ is of type $(1, \mathbb{Z}/\ell\mathbb{Z})_{\pm}$. Since $\ker\psi = \{q^r \bmod q^{\ell\mathbb{Z}} : r \in \mathbb{Z}/\ell\}$ and $\pi_W(q^{1/\ell}) = q$, the localization of $\underline{\epsilon}$ is the cusp of the generator ± 1: the label counts deck translations by the period q, and no Tate order or ramification index enters it. The rational 2-torsion gives "$K = \ddot{K}$", and the Weil pairings give $\mu_{60} \subset F$ and $\mu_\ell \subset K$.

4.7 The hypotheses of [IUT4, Thm. 1.10]

[S] [IUT4, Thm. 1.10] is stated for initial Θ-data in the situation of [IUT3, Cor. 3.12]. It assumes that E_F has good reduction at every nonarchimedean place of F over $\mathbb{V}_{\mathrm{mod}} \setminus \mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ that does not divide 2ℓ, and that $F = F_{\mathrm{mod}}(\sqrt{-1}, E_{F_{\mathrm{mod}}}[2 \cdot 3 \cdot 5])$; that is, F is obtained from the field $F_{\mathrm{tpd}} = F_{\mathrm{mod}}(E_{F_{\mathrm{mod}}}[2])$ by adjoining $\sqrt{-1}$ and the $(3 \cdot 5)$-torsion of the model of E over F_{tpd} given by a Legendre equation.

[P] Every place of L of odd residue characteristic at which E has bad reduction is in $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ (§4.1), so the first assumption holds. For the second, $E[2] \subseteq E(L)$, so $F_{\mathrm{tpd}} = L$; $\sqrt{-1} \in L$; (1) is a Legendre equation over L; and $F = L(E[15])$.

5 The levels and condition (C1) of [IUT4, Cor. 2.2]

[S] [IUT4, Cor. 2.2(i)] defines $\log(\mathfrak{q}^{\forall})$ as the normalized degree of the effective arithmetic divisor determined by the q-parameters of the curve at all nonarchimedean places. The initial Θ-data constructed in [IUT4, Cor. 2.2(ii)] have a prime ℓ with

$$\text{(C1)} \qquad (\log \mathfrak{q}^{\forall})^{1/2} \le \ell \le 10\delta \cdot (\log \mathfrak{q}^{\forall})^{1/2} \cdot \log(2\delta \cdot \log \mathfrak{q}^{\forall}), \qquad \delta = 2^{12} \cdot 3^3 \cdot 5 \cdot d,$$

where d bounds the degree of the point $x_E \in U_X(\overline{\mathbb{Q}})^{\le d}$ of the λ-line.

[P] Over F the curve is semistable, and at a place w of multiplicative reduction $\mathrm{ord}_w(q) = -\mathrm{ord}_w(j)$. Since the normalized degree does not change under extension of the base field [IUT4, Def. 1.9(i)], the sum can be taken over the nonarchimedean places v of L: $\log \mathfrak{q}^{\forall} =$

$\frac{1}{2}\sum_v \max(0, -\operatorname{ord}_v j)\log \mathrm{N}\, v$. The contributions are 944 at v_π, $2\operatorname{ord}_{\mathfrak{p}}(1-a)$ at an odd place $\mathfrak{p} \mid 1-a$, and 2 at the place above 2, where $\operatorname{ord}_{1+i}(j) = 3\cdot 8 - (8 + 2\cdot 9) = -2$. So

$$\log \mathfrak{q}^{\forall} = 472\log 2357 + \log \mathrm{N}(1-a) - 8\log 2 = 7324.751610927\ldots,$$
$$(\log \mathfrak{q}^{\forall})^{1/2} = 85.584762726\ldots,$$

and Λ is exactly the set of primes in $[(\log \mathfrak{q}^{\forall})^{1/2},\ 472/3)$. With $d = 2 = [\mathbb{Q}(a) : \mathbb{Q}]$ the upper bound of (C1) is about 2.2251×10^{10}, and it only grows with d. So every $\ell \in \Lambda$ satisfies (C1).

Remark 5.1. [P] The upper end 472/3 of the window is not part of (C1) or of [IUT1, Def. 3.1]. [S] It is the threshold $\ell < N/3$ of [Second, §§8,13], where $N = 472$, at which the margin $\operatorname{ord}_{2357}(\underline{q}_{\mathrm{b}}) - 3 = (472 - 3\ell)/\ell$ of the first stage is still positive; it is 1/157 at $\ell = 157$. [IUT4, Cor. 2.2] also requires that x_E lie in a compactly bounded subset satisfying the condition $(*_{j\text{-inv}})$ at 2 and outside a finite exceptional set $\mathfrak{Exc}_d$, which is not given explicitly. [P] These requirements are not checked here; (C1) is the only condition of [IUT4, Cor. 2.2] compared with the data.

6 Reproduction

- In the archive deposited with this note, `scripts/d2357_check.py` (Python 3, standard library only, about ten seconds) recomputes the numbers of Proposition 3.2 — for part (d), the valuations at the 277,462 places of the 138,731 primes; the reduction to these places is Lemma 3.1 and the case analysis in the proof — as well as Table 1, Table 2 and the remarks after it, the local invariants of §4.4 and the values of §5, and compares them with the numbers printed here. It also compares the reduction of a modulo every odd prime below 3000, $p \neq 2357$, with direct modular exponentiation. Its output is `outputs/d2357_check.txt`; it exits with a nonzero status if a comparison fails.
- The second computation of Remark 3.3 is `scripts/powerfree_certificate.py` in the deposit of [Second]; run unchanged, it reproduces the counts stated there.
- The arguments that are not computations — Lemma 3.1 and the reduction in the proof of Proposition 3.2(d), the image argument of §4.3 apart from the point counts and the residues, the stable and the split reduction, the core, the choice of places and cusp, and §4.7 — are not checked by these programs.

Use of generative AI

Generative AI tools (large language models) were used extensively in this work, under the author's direction, in finding the example and its proofs, in the computations and in the writing. The author takes full responsibility for the contents of this report.

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